

SOUMILI SARKAR

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PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate with hands-on experience in Machine Learning, Deep Learning, and Full-Stack Development. Proven track record building end-to-end AI/ML pipelines using TensorFlow, Python, and NLP techniques. Certified in Generative AI and Data Science with a strong CGPA of 9.06/10. Seeking a Software Engineer / AI-ML Engineer / Data Analyst role to deliver measurable business impact through intelligent systems.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++, SQL

ML / AI Frameworks: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, NLP (NLTK / spaCy)

Data Visualization: Matplotlib, Seaborn, Plotly, Streamlit, Power BI, Tableau

Web & Frontend: React.js, Next.js, HTML5, CSS3, REST APIs

Databases & Tools: MySQL, Git/GitHub, VS Code, Jupyter Notebook, Google Colab

Core Competencies: Data Preprocessing, Feature Engineering, Model Evaluation, Data Visualization, BI Reporting, Agile, Problem-Solving

PROJECTS

Diabetes Prediction System using ANN | *Personal Project* | 2025

- Designed and trained a 3-layer Artificial Neural Network (ANN) using TensorFlow/Keras to predict diabetes risk from 8 clinical features, achieving ~82% classification accuracy on the Pima Indians dataset.
- Engineered a preprocessing pipeline with Pandas and NumPy, handling missing values and normalizing features — reducing model error by ~15% compared to raw input baseline.
- Implemented probability-based output with Matplotlib visualizations and deployed an interactive prediction interface using Streamlit, improving prediction interpretability for non-technical stakeholders.

Mental Health Detection & Trend Analysis System | *Group Project* | 2025

- Collaborated in a team to build an NLP-powered machine learning model classifying mental health conditions from textual data, applying tokenization, stopword removal, and TF-IDF feature extraction using NLTK.
- Led trend analysis module — performed time-series analysis on 5,000+ records, surfacing behavioral patterns with interactive Seaborn and Plotly dashboards, enabling data-driven insight generation.
- Coordinated model evaluation and reporting across team members, improving interpretability using SHAP-style analytical insights and increasing overall classification confidence.

Calorie Tracker Web App with AI Image Analysis | *Personal Project* | 2025

- Developed a responsive full-stack web application using HTML5, CSS3, and JavaScript integrating the Grok API for real-time AI-powered food image recognition and calorie estimation.
- Implemented drag-and-drop and live camera capture UI features, reducing food logging time by ~60% compared to manual entry; integrated ImgBB API for seamless image hosting.
- Designed a modern, mobile-first interface with zero external CSS frameworks, achieving sub-2s load time and 100% client-side processing.

INTERNSHIPS & TRAINING

Industrial Training — Generative AI | *Ardent Computech Pvt. Ltd.* | Feb 2026 – Mar 2026

- Completed intensive training on Generative AI concepts including LLMs, prompt engineering, and fine-tuning techniques; applied learnings to build a prototype chatbot using Python.

Industrial Training — AI & Machine Learning | *EduSkills Foundation* | Jul 2025 – Sep 2025

- Completed 300+ hours of hands-on AI/ML training covering supervised learning, neural networks, and model deployment; implemented 5+ end-to-end ML projects in Python.
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EDUCATION

B.Tech in Computer Science & Engineering | *Narula Institute of Technology, Kolkata* | Aug 2023 – Present
CGPA: 9.06 / 10.0 (up to 5th Semester)

ISC Class XII (Science) | *Pearls of God School, Hooghly* | 2021 – 2023
Percentage: 82%

CERTIFICATIONS & ACHIEVEMENTS

- Programming in Java — NPTEL (Score: 94%) | Nov 2024
- Introduction to Data Science — Certified | Mar 2026
- Young Professional Course — TCS iON (AI, Business Etiquette, Communication) | Jul 2025
- Technology Job Simulation — Deloitte (Software Development & Python) | Jun 2025
- Certificate of Merit — Naukri Campus Young Turks 2025 (96.27 percentile) | Oct 2025